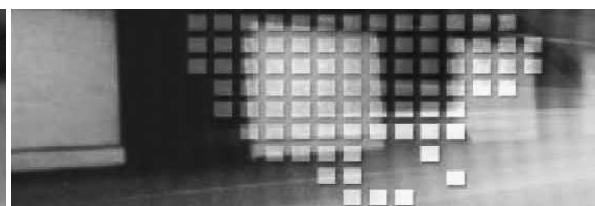
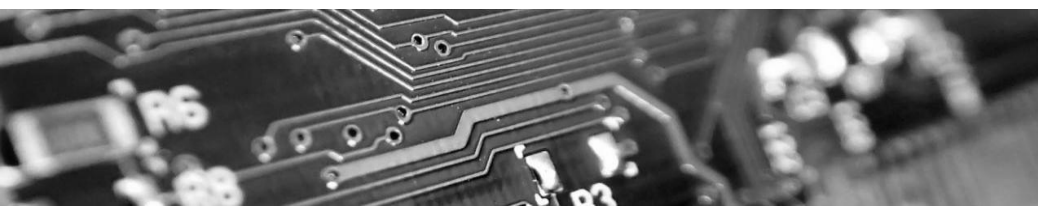




Home Automation & Power Control Instructions Manual



Version:
I

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INSTRUCTIONS MANUAL
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Introduction

The main goal of this instruction manual is to create a set of steps that will easily guide the user how to build a network, in a home or industry, to link electronic devices to save energy and money while protecting all electronic devices within the network. First, the home automation application will be able to monitor the amount of energy that users consume by placing a ZigBee module into our daily used devices such as lamp, radiator, ventilation, and others. Second, a smart meter will monitor every electrical device and combine them to form a large and complex network within a home or industry. In this way, the smart meter will shut any device off when needed as well as when too much energy is used. Third, the user will be able to manage their consumption conveniently and in detail. Finally, our set of instructions to build this network will not only make life more comfortable but also save energy and money while protecting our own environment.

1.0 GENERAL INFORMATION

GENERAL INFORMATION

1.1 System Overview

- System environment
 - This network set up is affordable and friendly to the environment.
- ZigBee Network Technology
 - It is one of the simplest and cheapest technologies with the best innovation in automation.
- Conserve and Control Power
 - It conserves electronic devices consumed energy while protecting them from overloading.
- Home Automation & Power Control
 - Name assigned to the team in charge of this network setup's instruction manual.
- Coordinator, Routers, and End Devices
 - Main Components needed to build a network with ZigBee technology.

1.2 Organization of the Manual

Coordinator:

The coordinator maintains overall network knowledge. It is the most sophisticated of the three types and requires the most memory and computing power.

Routers:

As well as running an application function, a router can act as an intermediate router, passing on data from other devices.

End Devices:

Contain just enough functionality to talk to the parent node (either the coordinator or a router, and they cannot relay data from other devices. This relationship allows the node to be asleep a significant amount of the time thereby giving long battery life.

2.0 SYSTEM SUMMARY

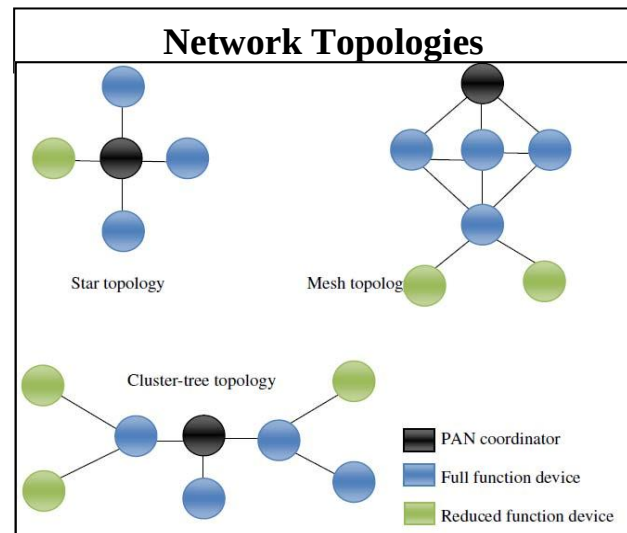
SYSTEM SUMMARY

2.1 System Configuration

ZigBee technology has three different network topology configurations: star, tree, and mesh topologies.

In a star topology, the network controlled by one single device is the ZigBee coordinator. All other devices, known as end devices, directly communicate with the coordinator.

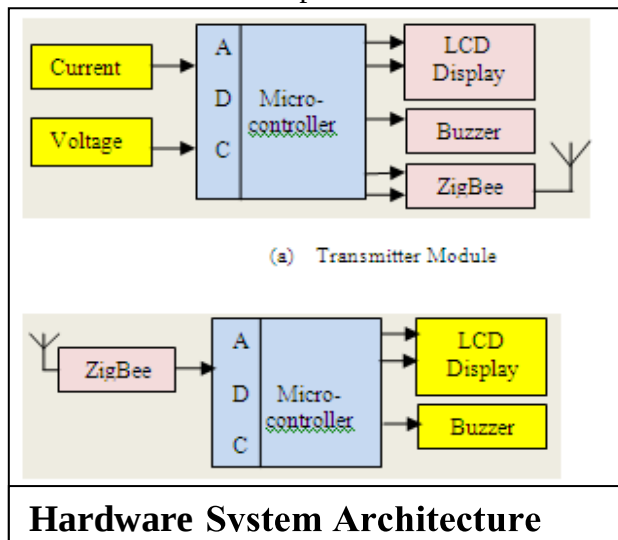
In mesh and tree topologies, the network may be extended with ZigBee routers. Mesh networks allow full peer-to-peer communication. This specification describes only



intra-PAN networks, that is, networks in which communications begin and terminate within the same network

Looking to the need of trustworthy power monitoring system, a ZigBee modeled system has to be comprised of the following modules:
Transmitter module
Transceiver module

The setup consists of the following elements; end devices, microcontroller, LCD display, and ZigBee module. Microcontroller mainly means for the function of the data processing, data storage, human interfacing and interoperability with the external environment.



- To choose requirement specific microcontroller with further scope of up gradation
- To study development tools with their version specification
- To study and apply High Level Language specification for application development

For this network setup, we chose the controllers and developments tools of Microchip Technology. Module of XB PRO Series2 of DIGI will be used here as a part of ZigBee stack which will then be addressed by mixed API mode with hardware controlled flow control for USART interfacing with the desired baud rate. ZigBee protocol is for wireless communication between transmitter and transceiver module. LCD display displays the current and voltage parameters.

3.0 GETTING STARTED

GETTING STARTED

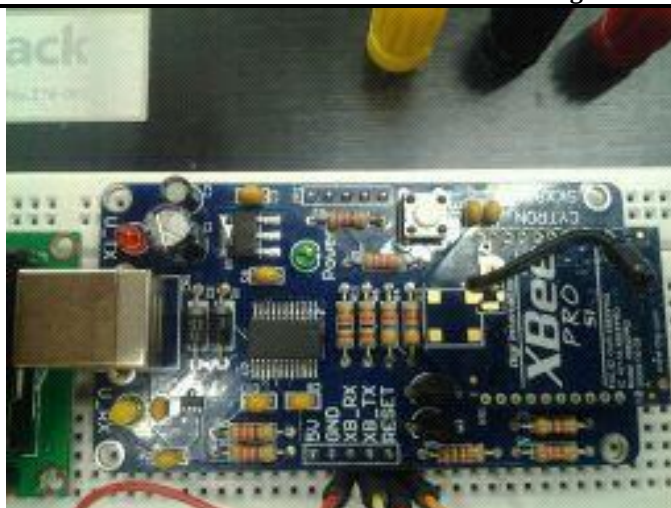
a. Connect USB cable (B type) to the SKXBee USB socket, another end (A type) to a PC as shown in following figure.

b. Follow driver installation guide.

c. SKXBee is ready, once plugged into the PC.

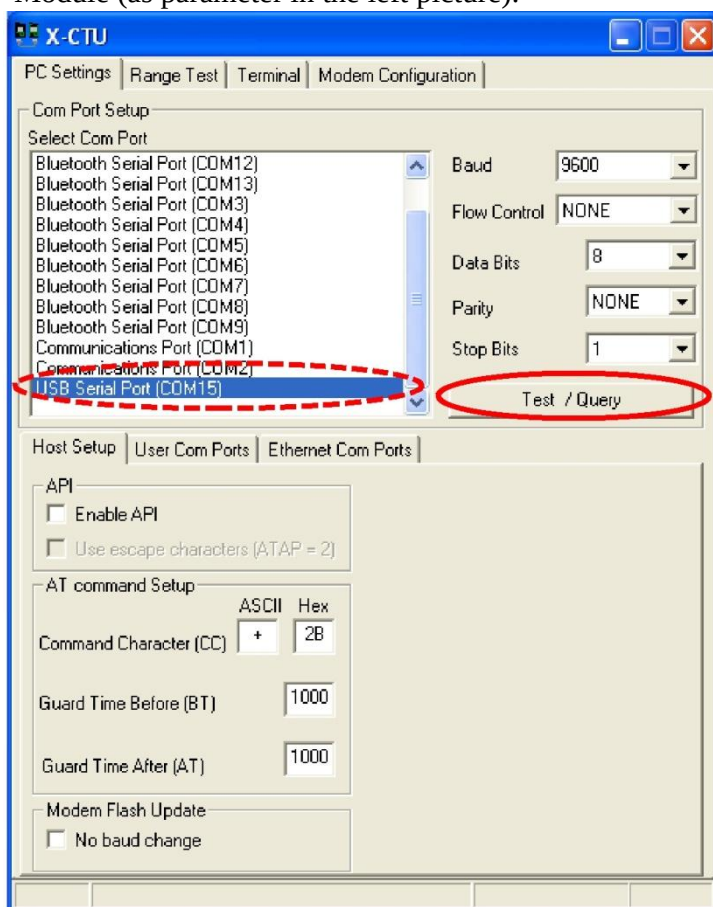
d. Use X-CTU, software used to communicate with XBee, from Digi International.

First, setup X-CTU to change configuration or transmit data.



Install the X-CTU Software to the PC by double clicking on “setup_x-ctu.exe” file downloaded from the SKXBee website. X-CTU is configured to enable the communication and display data from XBee module. Please follow the instruction as stated below to setup the computer.

e. In X-CTU Software click on “PC Settings” tab. Verify the settings to match those in XBee Module (as parameter in the left picture).

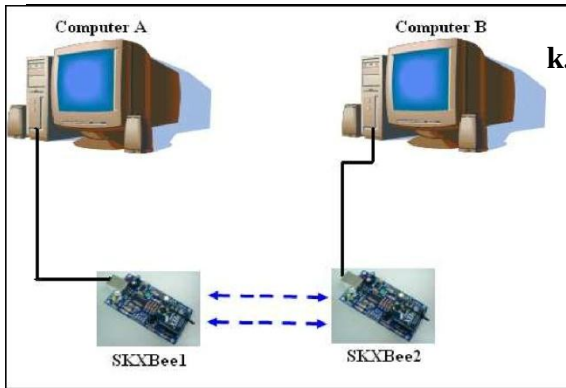


f. Choose the COM Port and click ‘Test/Query’ tab. Then, click ‘OK’.

g. To modify or read XBee Module parameters use AT command Mode.

h. First, select the ‘Terminal’ tab on X-CTU Software. Enter AT Command Mode and send the 3-character command sequence ‘+++’ and XBee module will response ‘OK’ which means that it has already enter Command Mode.

i. If a command is successful the module will returns an “OK” message, otherwise the module will return an “ERROR” message.



k. To exit AT Command Mode, send ATCN Command. CT (Command Mode Timeout) Command, the XBee Module automatically returns to Idle Mode. Therefore, if another command needs to be sent you must return to Command Mode.

l. To communicate between two SKXBee, example TRANSMITTER and TRANSCIEVER, Source Address (MY) of both XBee1 and XBee2 must be match to Destination Address (DL) of both XBee1 and XBee2.

m. Set the address for each module as shown below.

SKXBee2 MY = 2222
DL = 1111

SKXBee1 : MY = 1111
DL = 2222

o. After building the connection between the two XBee they will communicate among each other.

p. Below is an example of data sent and received of XBee1 and XBee2 in X-CTU terminal.

```

X-CTU [COM15]
PC Settings | Range Test | Terminal | Modem Configuration
Line Status: [OK] [DSR] [DTR] [RTS] [Break]
Close Com Port Assemble Packet Clear Screen Show Hex

+++OK
atmy1111
OK
atwr
OK
atd12222
OK
atwr
OK
atcn
OK

+++OK
atmy
1111
atd1
2222
atcn
OK
  
```

```

X-CTU [COM15]
PC Settings | Range Test | Terminal | Modem Configuration
Line Status: [OK] [DSR] [DTR] [RTS] [Break]
Close Com Port Assemble Packet Clear Screen Show Hex

+++OK
atmy2222
OK
atwr
OK
atd11111
OK
atwr
OK
atcn
OK

+++OK
atmy
2222
atd1
1111
atcn
OK
  
```

SKXBee1	SKXBee2
<pre> X-CTU [COM15] PC Settings Range Test Terminal Modem Configuration Line Status: [OK] [DSR] [DTR] [RTS] [Break] Close Com Port Assemble Packet Clear Screen Show Hex KASDKASLD JASDASD ESADASD YUUYU HGUYUYUY YU9IU00 HUIUIUOUP DSFDSFDGG TYTUUUUUU YGFUYUYTYU IOPOIPIOP UIOUIOUIO IPIOPIPIO IUIOUIOUP IOIUIOUIP IUOUI90-5465463 1524645456 414242 45454 545 45454 5 45 454 5 454 54 UYGIUYOYUO UYHIUYOUI </pre>	<pre> X-CTU [COM14] PC Settings Range Test Terminal Modem Configuration Line Status: [OK] [DSR] [DTR] [RTS] [Break] Close Com Port Assemble Packet Clear Screen Show Hex KASDKASLD JASDASD ESADASD YUUYU HGUYUYUY YU9IU00 HUIUIUOUP DSFDSFDGG TYTUUUUUU YGFUYUYTYU IOPOIPIOP UIOUIOUIO IPIOPIPIO IUIOUIOUP IOIUIOUIP IUOUI90-5465463 1524645456 414242 45454 545 45454 5 45 454 5 454 54 UYGIUYOYUO UYHIUYOUI </pre>

SKXBee1

SKXBee2

r. To restore XBee Module defaults setting, click 'Restore' button.

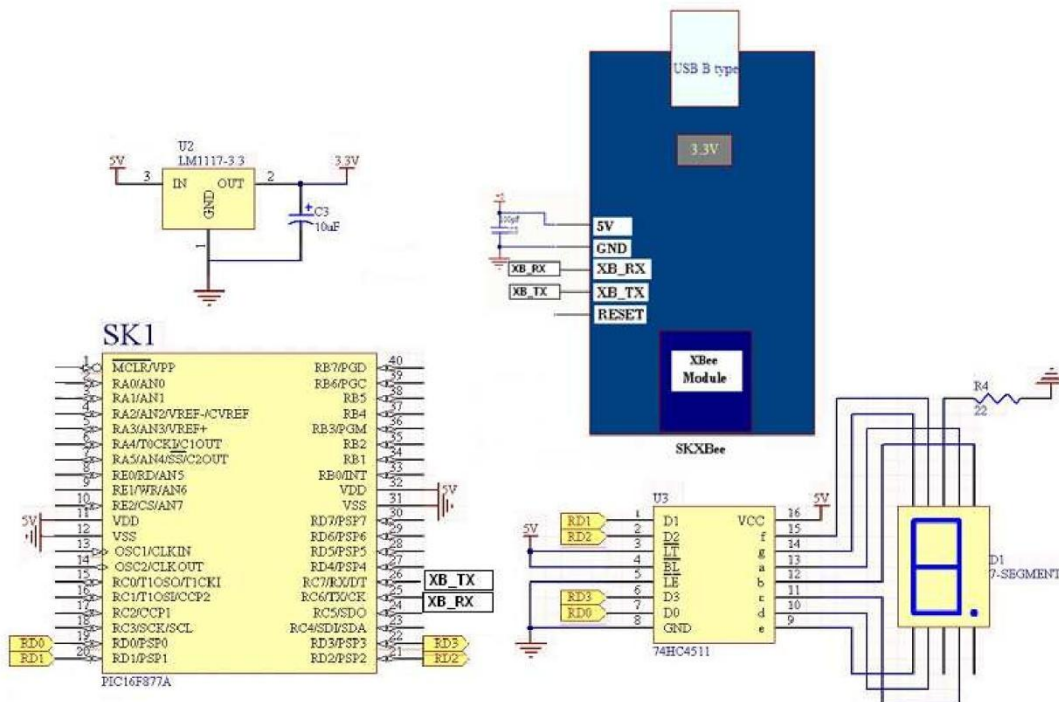
s. To modify SKXBee parameters refer to Command Reference Tables in Data Sheet; you can modify them by using ATCT (Command Mode Timeout).

4.0 SYSTEM PROCEDURES

4.1 Input Procedures and Expected Output

How to set up the SK40C and PIC as a coordinator and router simulator

- a. Setup SK40C with PIC.
 - Prepare a PCB board and all the components required according to the schematic.



- b. Load programs to the microcontroller using a PCI interface.
Sample source codes are downloaded from the web.

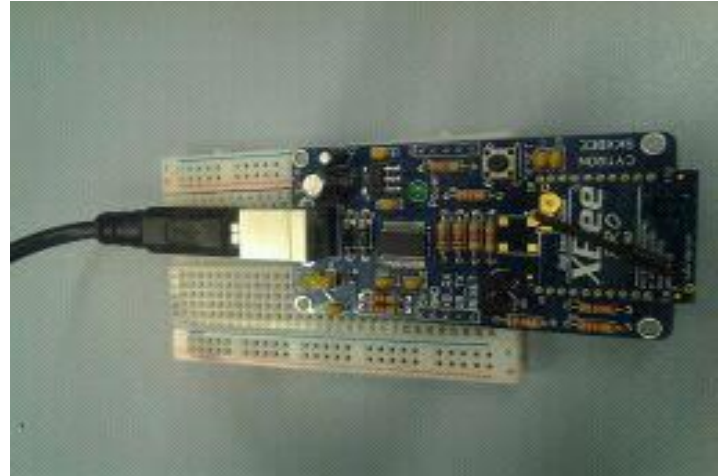
```
void init(void) // subroutine to initialize
{
    SPBRG=129; // set baud rate as 9600 baud
    BRGH=1;
    TXEN=1;
    CREN=1;
    SPEN=1;
    TRISD = 0b00000000;
    seg = 0b00000000;
}
```

- c. Turn on, press and release reset button of SK40C.
- d. After setting everything up, the SK40C will be able to receive data through the network.

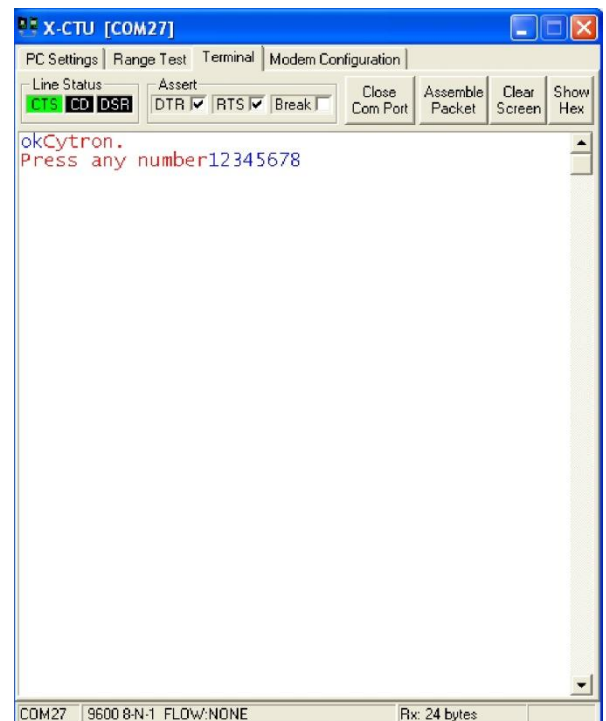
4.2 Using SKXBee as a Coordinator

Connect SKXBee to PC using USB cable as the left picture. Set the address for each module, test the communication and go to 'Terminal' setting tab.

b. Press reset button of SK40C, followed by pressing "ok" and "Press any number" will appear.



c. Connect end devices. Red color is for data received while blue color is for data sent.



d. Energy consumed will appear in the LCD screen.



5.0 Instruction Manual References

Works Cited

pheysia. (2011, July 30). *Cytron Technologies Official Website*. Retrieved March 03, 2012, from XBee/XBee Pro Wireless Communicate with Microcontroller: <http://tutorial.cytron.com.my/2011/07/30/xbeexbee-pro-wireless-communicate-with-microcontroller/>

Raut, Archana R., and L. G. Malik. (2011). ZigBee Based Industrial Automation Profile for Power Monitoring System. *International Journal On Computer Science & Engineering*, 3(5), 2028-2033.